

Proof of Concept and Pilot Purgatory

GLOSSARY 04 - SHEET 3 / 8

A bounded feasibility test, and the failure state in which pilots never reach production

PROFESSIONAL DEFINITION

A proof of concept is a small, time-boxed exercise designed to establish whether a proposed approach is technically feasible, and it is complete when that question is answered either way. Pilot purgatory is the organisational failure state in which a portfolio of such exercises accumulates, each judged successful on its own terms, while none reaches production - because the criterion applied was feasibility rather than scalability, and no exit decision was ever required (World Economic Forum and McKinsey, 2018).

WHO DEFINED IT ORIGINALLY

Proof of concept is a general engineering practice without a single originator. The pilot purgatory diagnosis was named by the World Economic Forum and McKinsey & Company in their Global Lighthouse Network research on Fourth Industrial Revolution manufacturing deployments, and has been elaborated in subsequent editions of that programme.

WHY THIS DEFINITION WAS DEVELOPED

The proof of concept is easy to approve, cheap, low-risk and locally rewarded, which produces a portfolio that grows without ever consuming the scarce resource that production deployment requires. Naming the accumulation as a distinct pathology made it a portfolio-level governance failure rather than a series of individually reasonable decisions.

FIGURE 1 - TWO EXITS FROM A PILOT

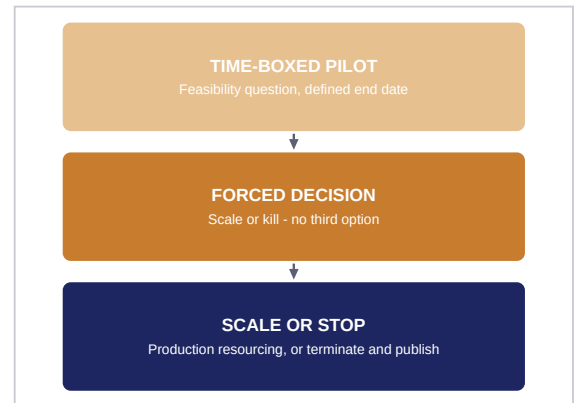


Figure 1. Purgatory is the absence of the middle block. Where no forced decision exists, a pilot neither scales nor stops; it persists at low cost, consumes attention, and is reported as evidence of progress (after World Economic Forum and McKinsey, 2018).

HOW IT IS USED

Define the exit criterion before starting - what result would cause this to proceed to production and what result would cause it to stop - and require an explicit kill or scale decision at the time box's end; a pilot with no defined end state does not terminate.

WHEN IT IS USED

At every pilot approval, at quarterly portfolio review, when pilot count exceeds production deployments, and when the same use case has been piloted more than once.

BY WHOM IT IS USED

Investment committees approving pilots, portfolio offices tracking conversion rates, business sponsors accountable for scaling, and the executives to whom pilot counts are reported as progress.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor governance instrument was the stage-gate, which permits progression but does not compel termination, so a pilot may remain indefinitely between gates. The purgatory framing adds the requirement of a forced decision at the time box and shifts the success criterion from technical feasibility - which almost every AI pilot satisfies - to scalability, which most do not. This changes what a pilot is for.

RELATED DEFINITIONS IN THIS COURSE

[AI Portfolio Management - Unit 3](#)
[AI Adoption - Unit 4](#)
[MLOps - Unit 4](#)
[Benefits Realisation - Unit 4](#)
[AI Roadmap - Unit 4](#)

SOURCES (HARVARD REFERENCING STYLE)

World Economic Forum and McKinsey & Company (2018) The Next Economic Growth Engine: Scaling Fourth Industrial Revolution Technologies in Production. Geneva: World Economic Forum.

World Economic Forum and McKinsey & Company (2019) Fourth Industrial Revolution: Beacons of Technology and Innovation in Manufacturing. Geneva: World Economic Forum.

Cooper, R.G. (1990) 'Stage-gate systems: a new tool for managing new products', Business Horizons, 33(3), pp. 44-54.

Verification note: figures on pilot-to-production conversion rates circulate widely in practitioner literature and vary substantially by source, sector and definition. Cross-check any specific percentage against the primary report and its stated methodology before using it in assessed work.